School to Work Pipeline for Water/Wastewater Industry: Update on the EdPASS-H2O EPA Grant

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USEPA Innovative Water Infrastructure Workforce Development Grant #84034501

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With special guest Nydia Lampkin



EdPASS-H2O Project Description

- K12 STEM-based curriculum framework
- Foundational knowledge and awareness:
  - water/wastewater issues
  - career opportunities/pathways



Grant Partnership:

ECU

NRWA

OKLAHOMA WATER SURVEY
OK H2O
at *The* UNIVERSITY *of* OKLAHOMA.

Goal 1: Produce water consumers who understand both the natural water cycle and the human water cycle













Goal 2: Produce water consumers who are aware of and actively work to alleviate current and future water issues



Goal 3: Produce water/wastewater infrastructure professionals prepared for this changing and challenging industry







- Enroll high school CareerTech students in local EdPASS-H2O pre-apprenticeship program
- Produce EdPASS-H2O high school graduates who qualify for the state RWA/NRWA Apprenticeship Program

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Partner Responsibilities







ECU and OU: EdPASS-H2O Roadmap

- Pre-apprenticeship *curriculum framework*
- Problem/project Based Learning Modules (P/PBL)
  - Supplemental cross-curricular *modules* aligned w/national educational standards
- *Simulation app* for use with modules and stand-alone
- Identify means of building relationships with local schools, Career Tech, and water/wastewater partners

NRWA/ORWA: SMEs and Support

- WaterPro Academy
  - House EdPASS-H2O pre-apprentice portfolios
  - Track pre-apprenticeship student progress

EdPASS-H2O Pre-Apprenticeship *Curriculum Framework*: **High School Knowledge**

	Min Educ Level	HS Math	HS Science
	Required	Required	Required
Water Systems Specialist **Wastewater Systems Specialist**	Water: HS Diploma or GED Wastewater: HS Diploma or GED required: 2- or 4-year degree recommended Licensure required • Multiple areas of content • Multiple levels • Must test for each level	Water: min math GPA 2.75 Wastewater: min math GPA 3.0 State minimum math course work required for HS graduation	Water: min science GPA 3.0 Wastewater: min science GPA 3.25 • Chemistry strongly suggested but not required State minimum science course work required for HS graduation - and - If not required by state min, include: • Environmental or Earth Science • Physical Science • Computer Science/Digital Tech

GPA Scale
A – 4.0
B – 3.0
C – 2.0
D – 1.0

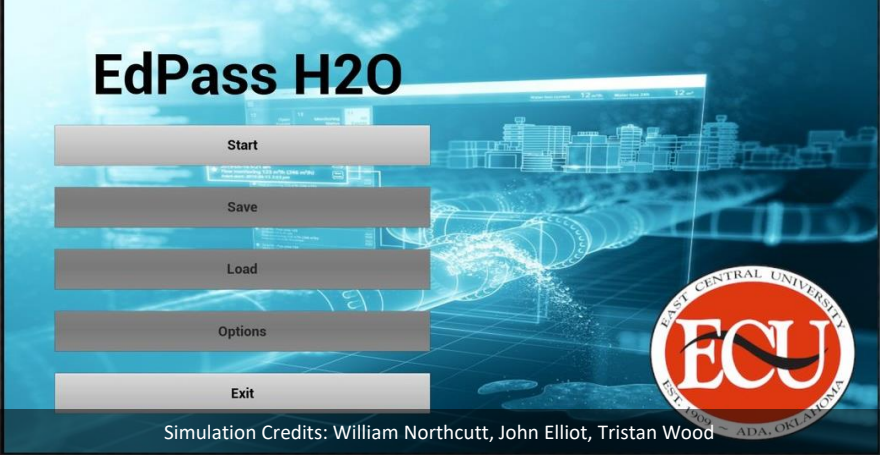
Outside of math/science, any recommended classes taken for EdPASS-H2O that are assigned a letter grade, min overall GPA 2.75

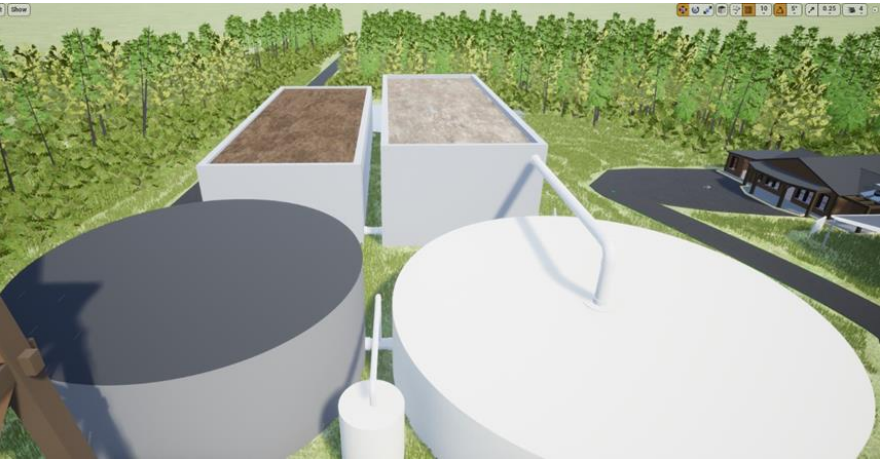
EdPASS-H2O Roadmap

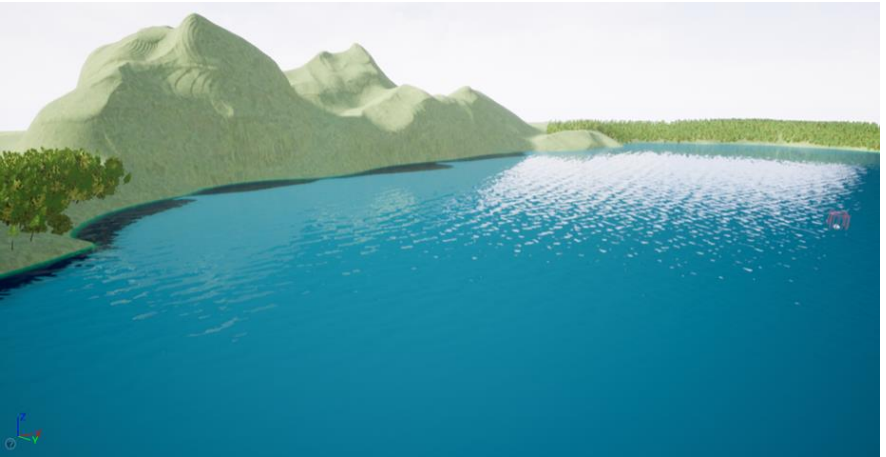
In addition to regular STEM curriculum, industry experts introduce students to water and the water/wastewater industry

4th - 6th grade	7th – 8/9th grade	9/10th – 12th grade
Identity Development	**Career Identity Development**	**Career Development**
• Local STEM fieldtrips • Invited STEM speakers	• Local STEM fieldtrips • Invited STEM speakers • Problem/project-based learning modules • Career curriculum & field trips	• Local STEM fieldtrips • Invited STEM speakers • P/pBL modules • Career curriculum & field trips • EdPASS-H2O Pre-Apprenticeship • CareerTech/High School





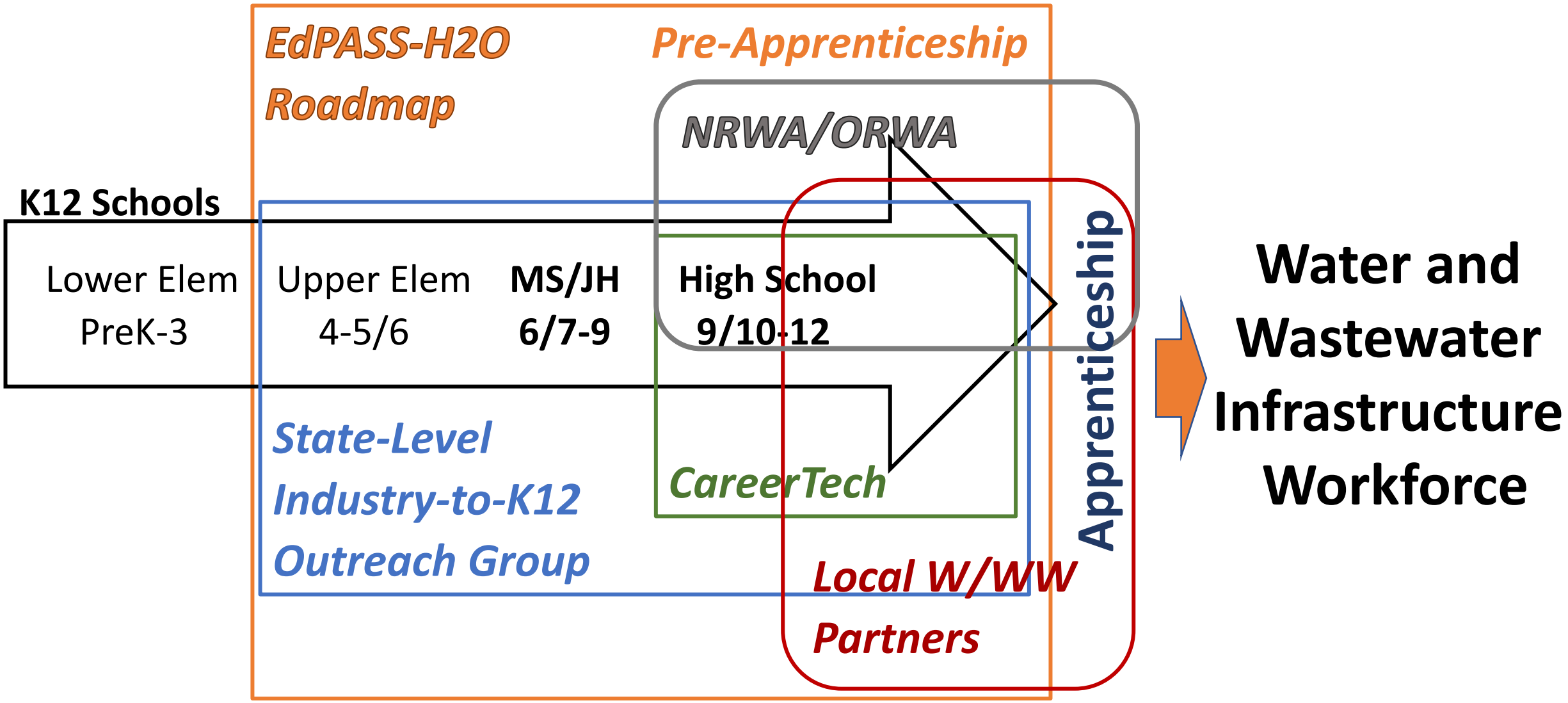




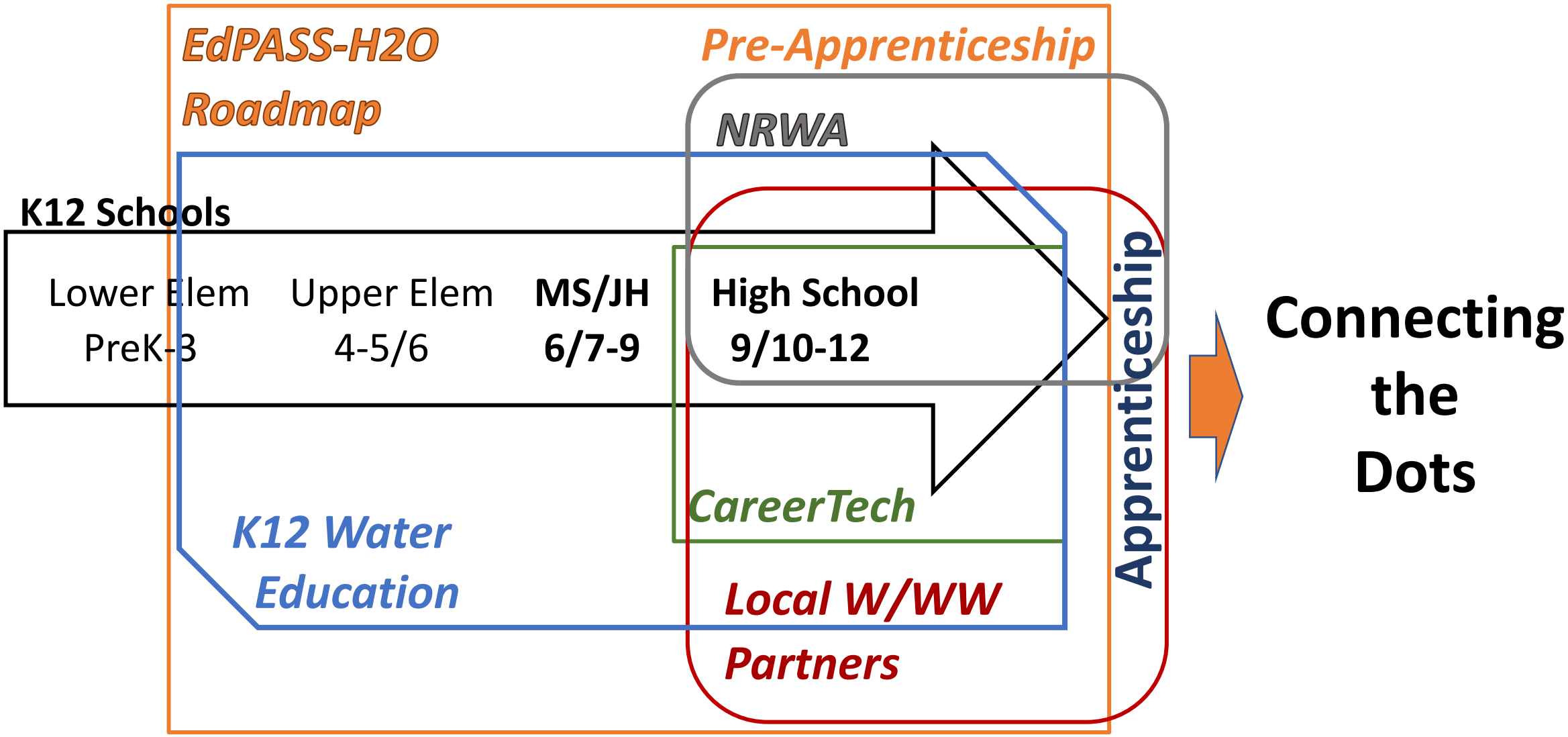
EdPASS-H2O Simulation

- Interactive game used to simulate work in the water industry
- Explore multiple environments
  - Water Operations Office
  - Wastewater Treatment Plant
  - Drinking Water Treatment
  - Pipes
  - Decentralized Water Treatments
  - Surface Water Locations

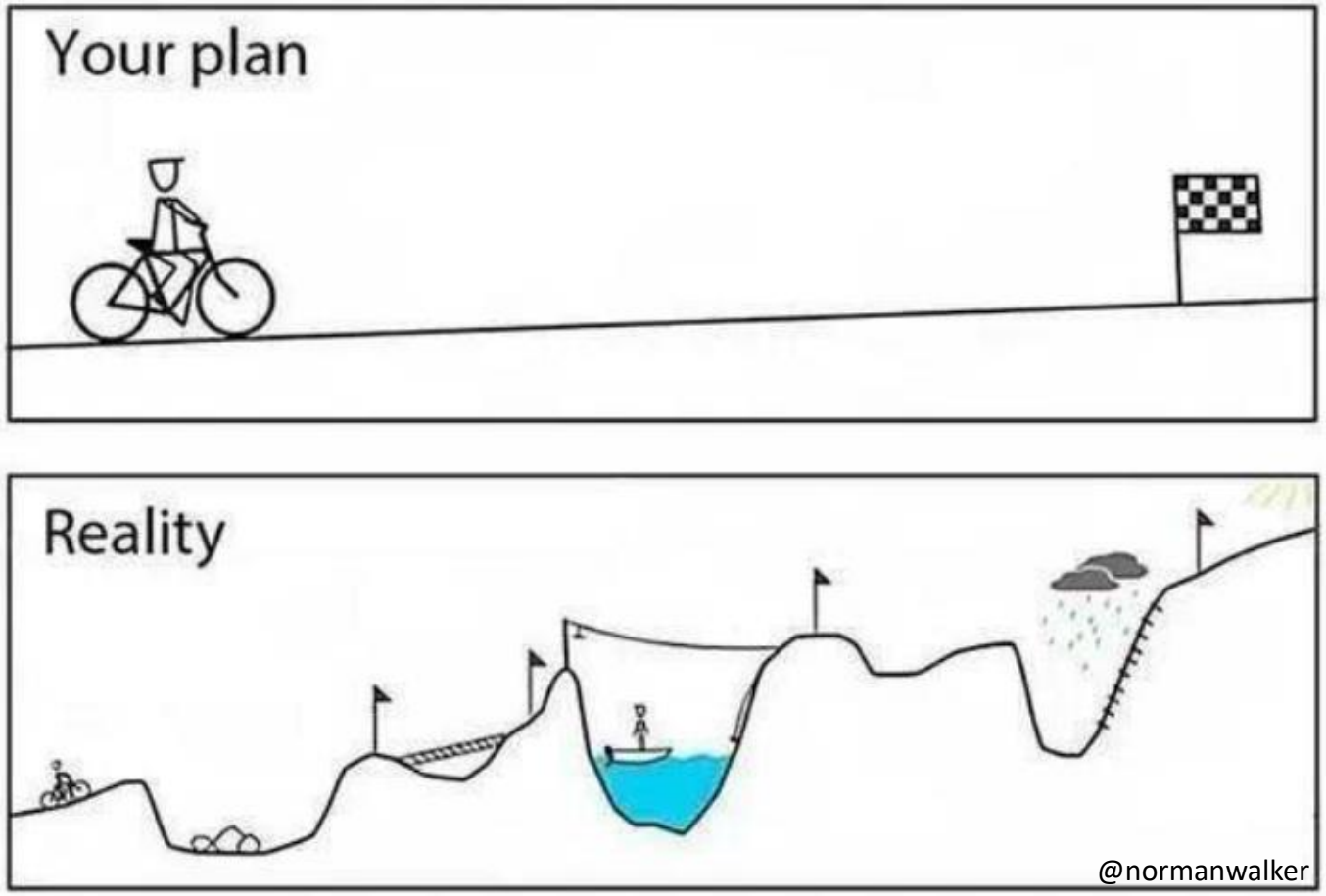
EdPASS-H2O Roadmap to Pre-Apprenticeship and Beyond



Current EdPASS-H2O Roadmap to Pre-Apprenticeship



Connecting the Dots…



Year 1

- Outreach
  - Tribal Nations
  - Local Rural Water Districts
  - Municipalities
  - Career Tech
  - State Level Entities
- Progress on Modules and Simulation App

Years 2 and 3

- Water-Focused Summer STEM Academies
- K12 Outreach
- Career Tech
- State and local Municipalities/RWDs
- K12 water education resources
  - Modules and Curriculum
  - Google Classroom
  - Videos and Links
  - Simulation App

Connecting the Dots…

Work with Career Tech

- Identified existing CT programs for *basic industrial and safety training*
  - Mechatronics
  - Welding
  - Industrial & Skilled Trades
- Work with CT counselors
  - Add water to careers discussion with students
  - Implement curriculum framework
  - Notify of state RWA apprenticeships

Work with State Entities and Local Municipalities/RWDs

- Identify municipalities and RWDs to partner with local Career Techs
  - Student job shadowing/pre-apprenticeships
- Career Outreach
  - OMAG Sanitary Sewer Expo
- Plant Tours
  - Hold Harmless/Liability Release
  - Caution List for K12 Teachers

PONTOTOC TECHNOLOGY CENTER
TRAINING TO SUCCEED

OKLAHOMA RURAL WATER ASSOCIATION
RWA



Connecting the Dots…

State-Level Industry-to-K12 Outreach Group

- State-level group or regional groups within the state
- Coordinated collaborative outreach efforts to K12
- Suggest working with local Career Tech, colleges, and universities to enhance or develop:
    - Local, regional, and state-level STEM events (ex. science fair)
    - STEM demo trailers for interactive roadshows (ex. aquifer model, watershed model, wetland model, valves/meters)
    - Field trips (ex. treatment plants, labs, water sources, infrastructure)
    - Summer camps (ex. STEM with water focus)





ScienceFest Oklahoma



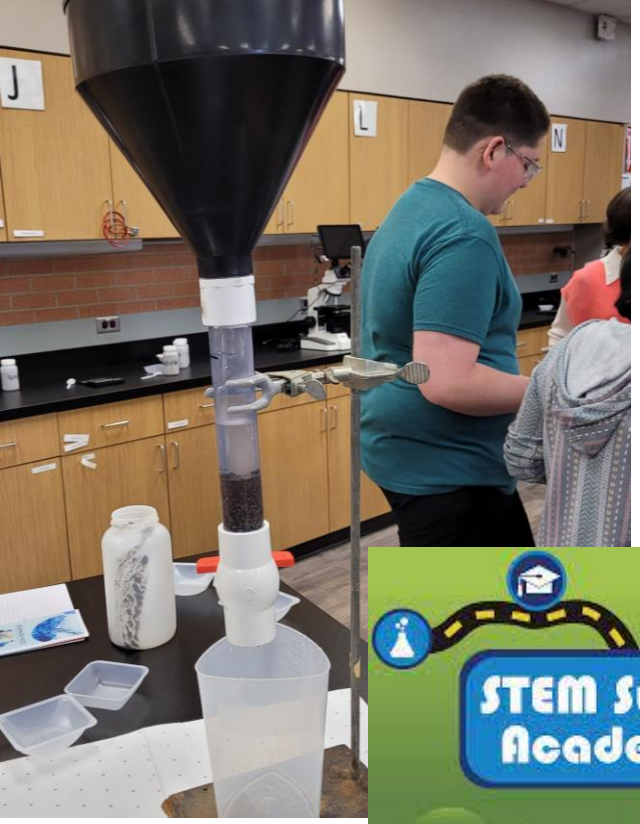


WaterPro Conference





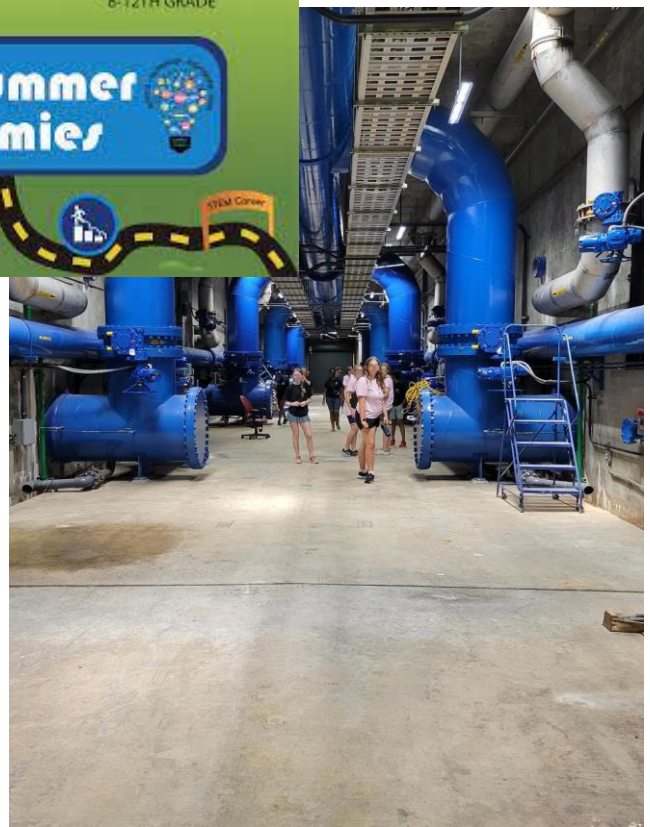












Last Summer: Oklahoma State Regents for Higher Education Summer STEM Academies

Students experience campus life and explore STEM career fields

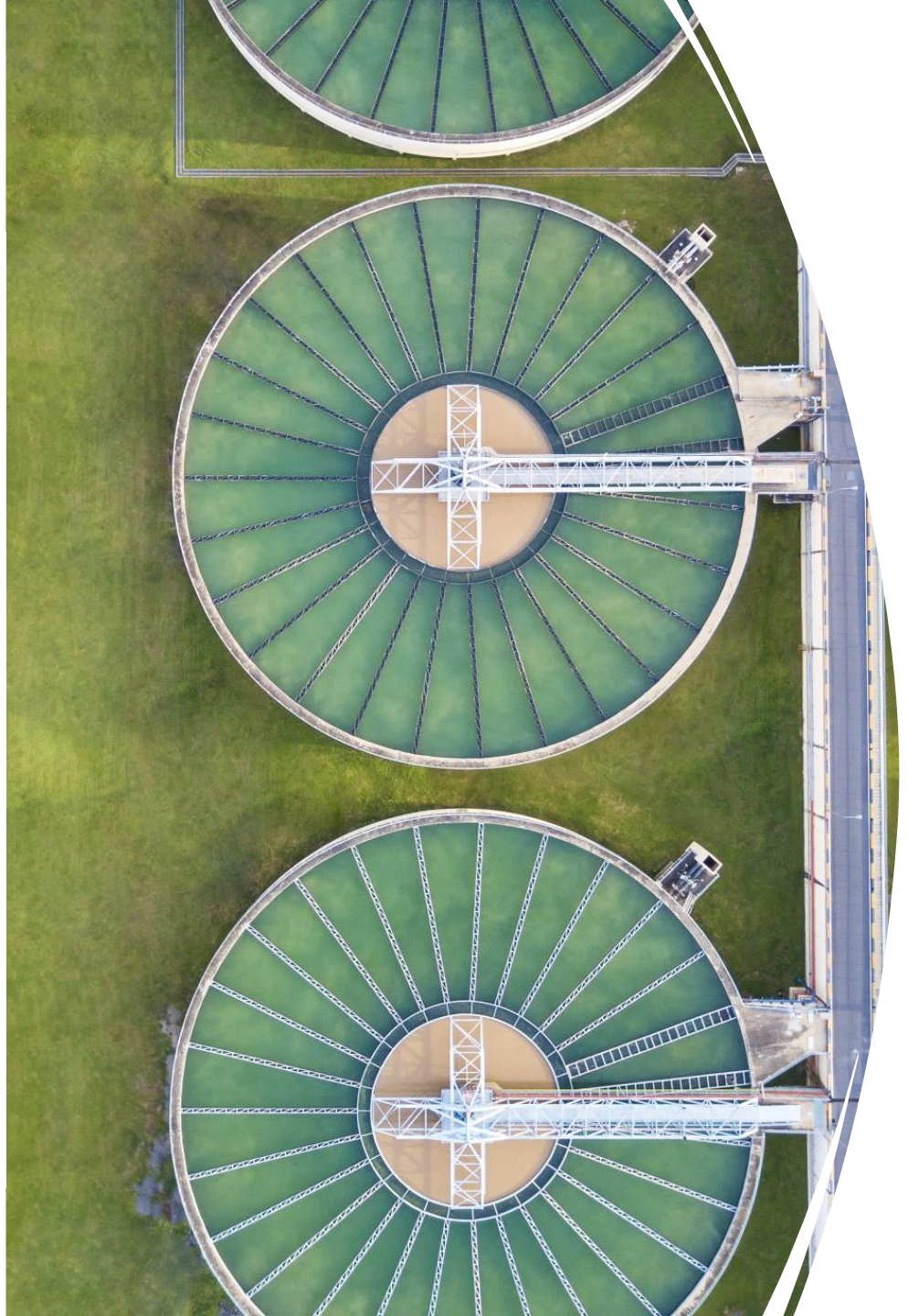
- Summer 2023 - visited 14 academies
- 2-4 hours each visit
- Multiple activities and labs:
  - Treatment plant tours
  - Microscopes
  - Coagulation and floccu lation
  - Filtration
  - Water quality test
  - Engineering challeng
  - Secchi disc build
  - Careers exploration
  - App simulation

Already invited back for next summer!

Future Plans…K12

Applied to host own Summer Academy

- The Human Water Cycle
  - If funded: 3 years & hosted at 3 sites/year
    - East Central University
    - University of Oklahoma
    - Rose State College
- Academy Modules
  - Water Quality and Science
  - Water Source Management & Sustainability/GIS
  - Drinking Water Treatment
  - Wastewater Treatment
  - Engineering Day
- *Will continue to visit other Summer STEM Academies*



Future Plans…K12

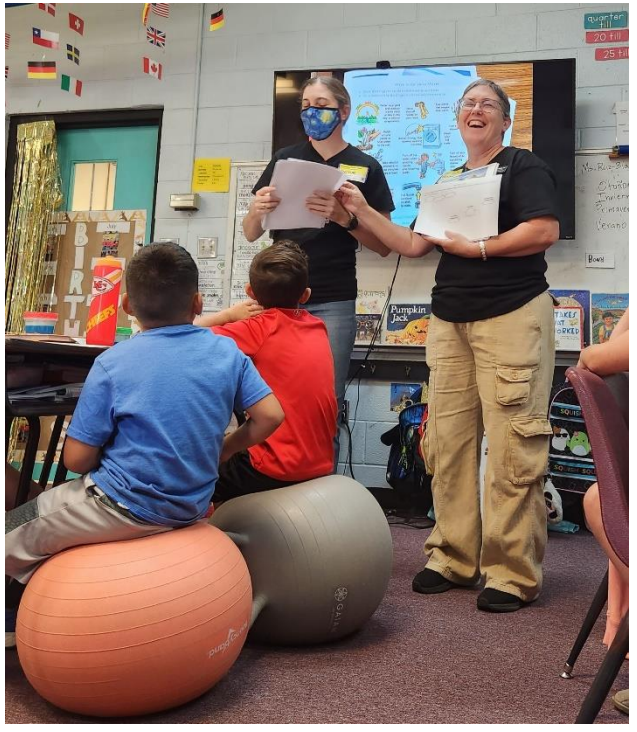
W is for Water

- Partnership with local K12
- Integrate water education
  - Natural and Human Water Cycle
  - Conservation/sustainability
  - Guest speakers
  - Class trips
- This year:
  - Fall: 3rd grade and AP Env Science
  - Spring: approach other local K12s
- Next year:
  - Expand grade levels





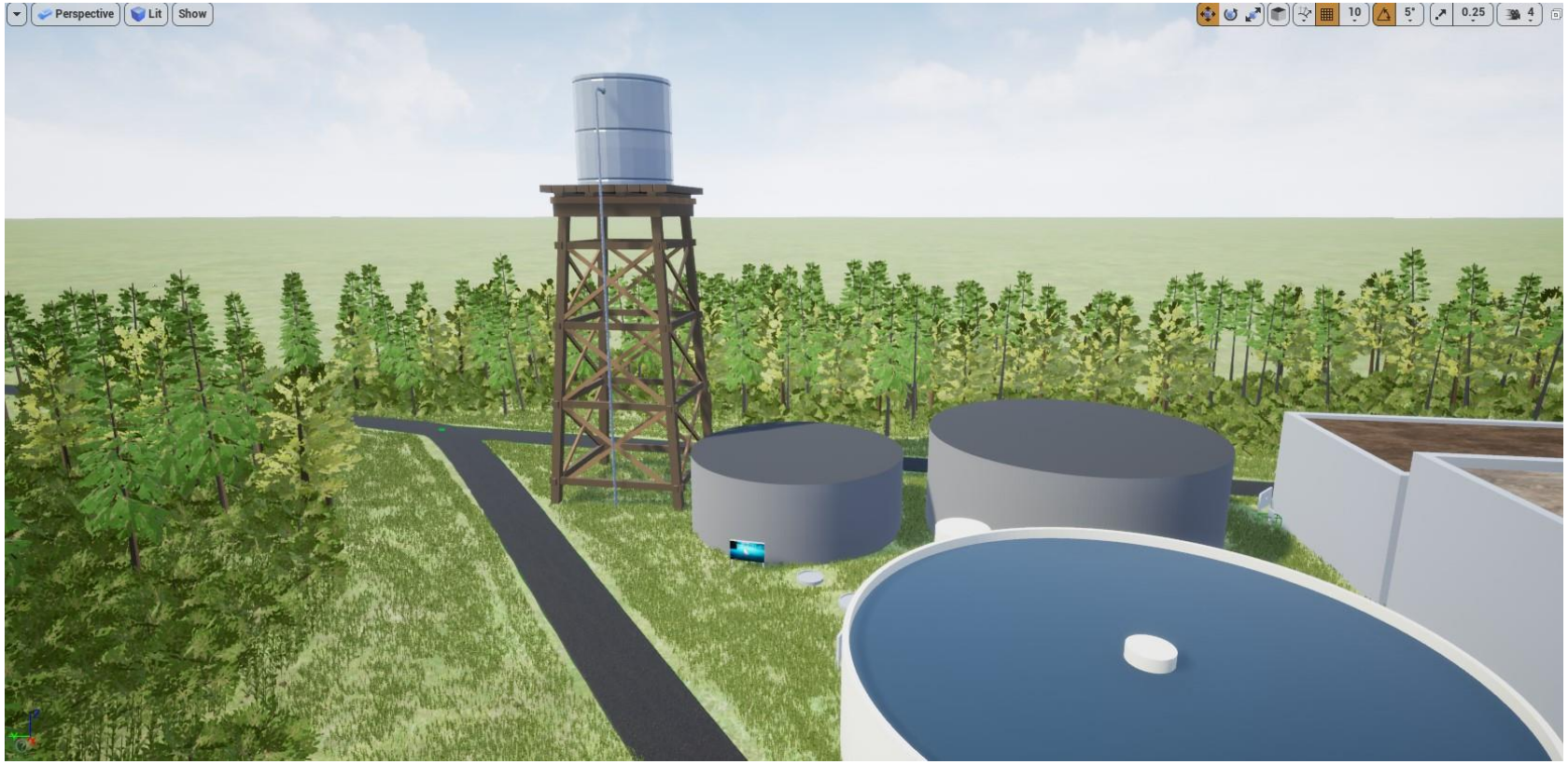




Future Plans…K12

Long Term Educational Resource Availability is a Priority

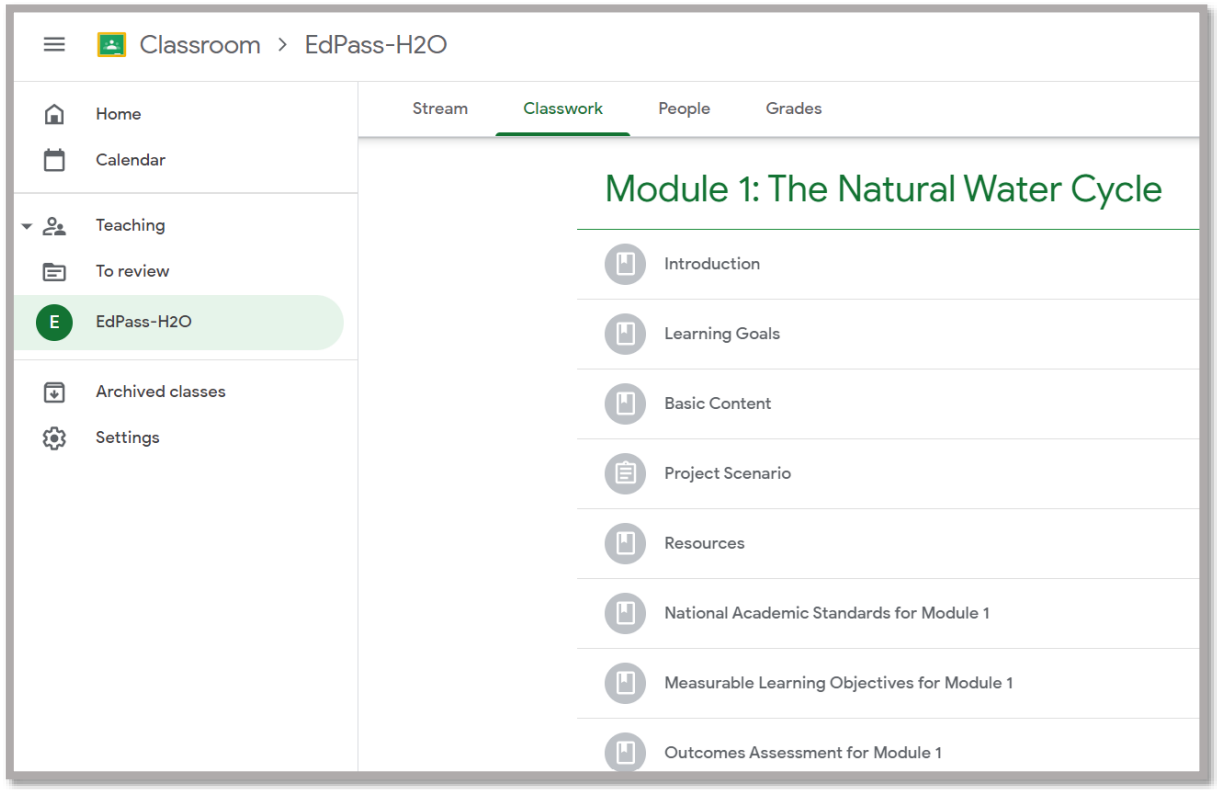
- Publicly accessible server for simulation

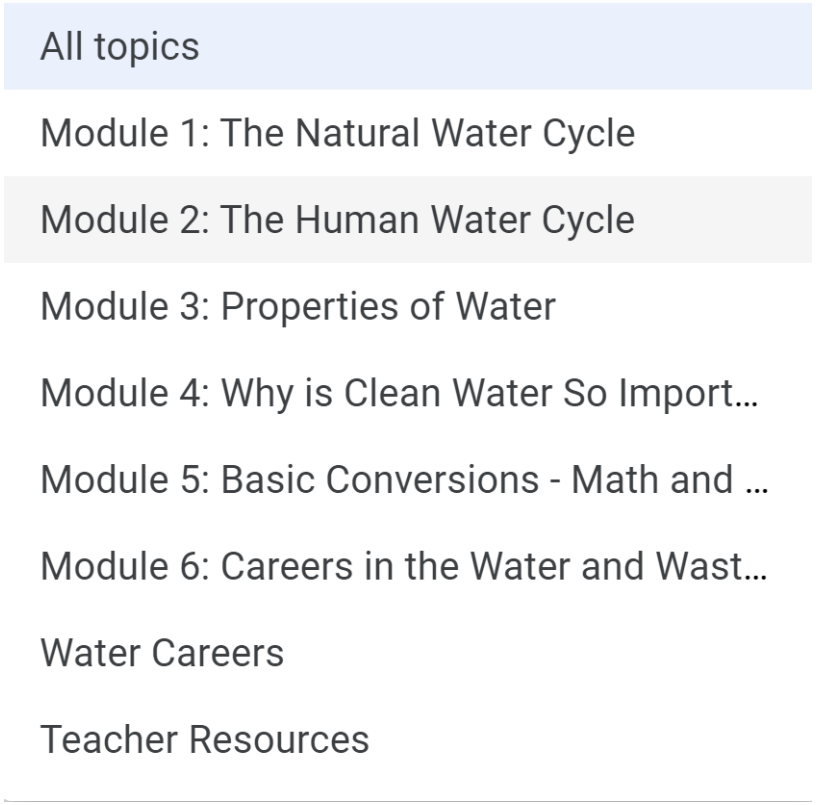


Future Plans…K12

Long Term Educational Resource Availability is a Priority

- Develop open-source curriculum using Google Classroom
  - Exportable for multiple applications

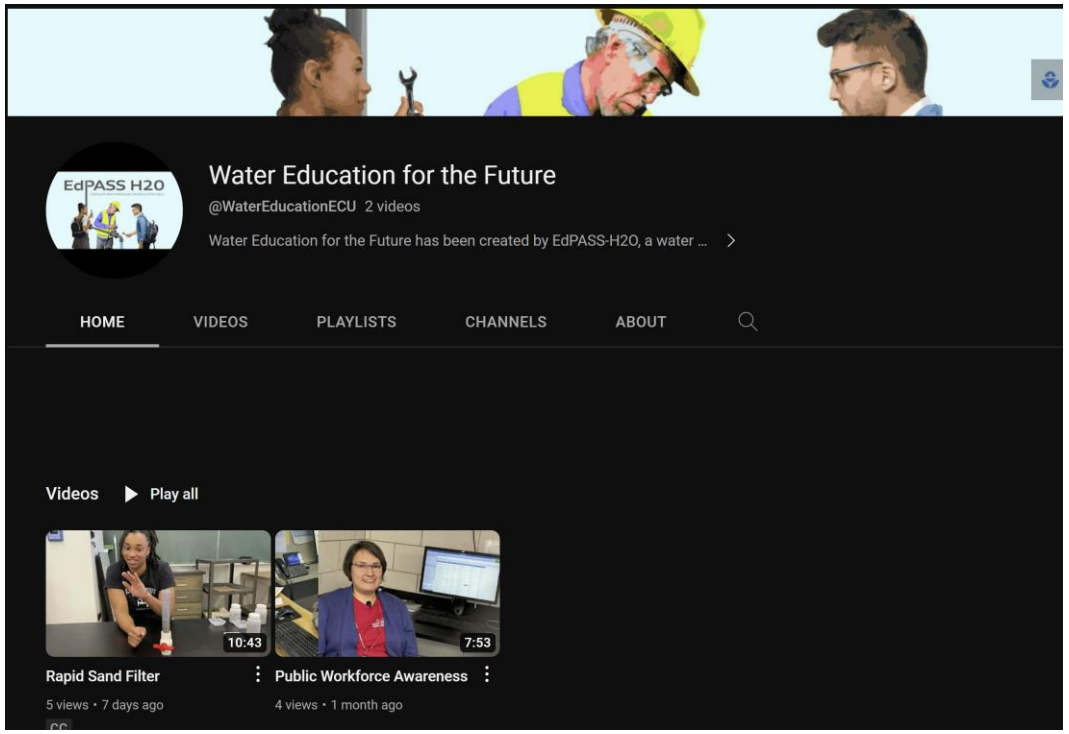


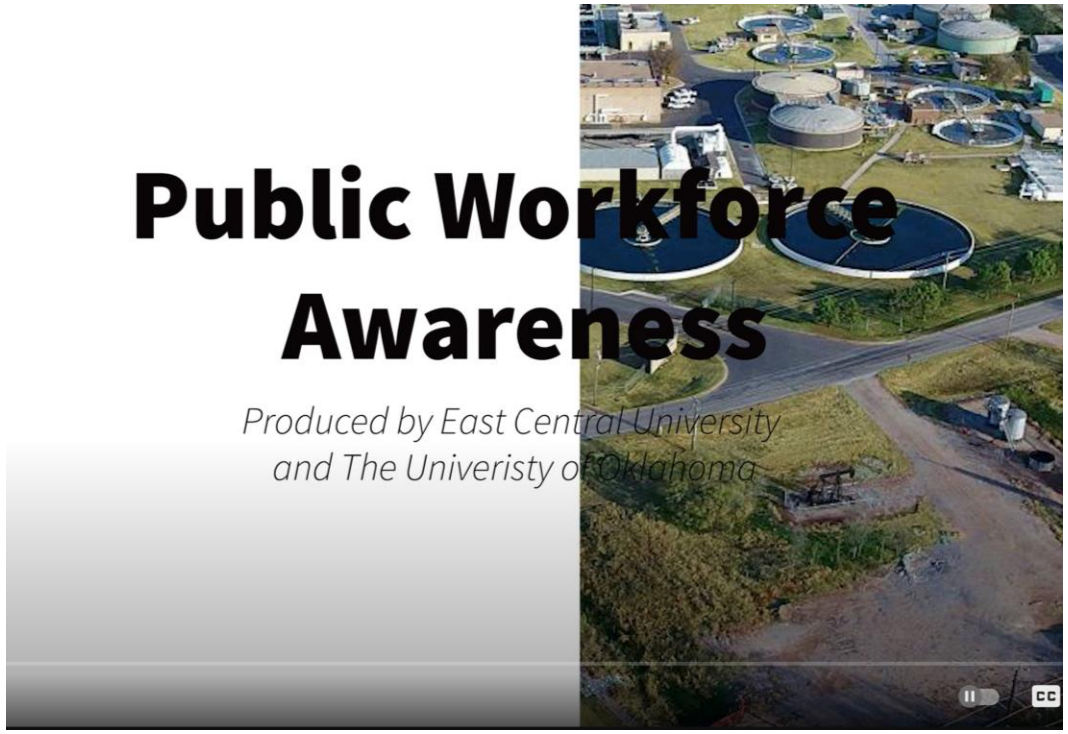


Future Plans…K12

Long Term Educational Resource Availability is a Priority

- www.youtube.com/@WaterEducationECU/





Teacher Accessible Activities



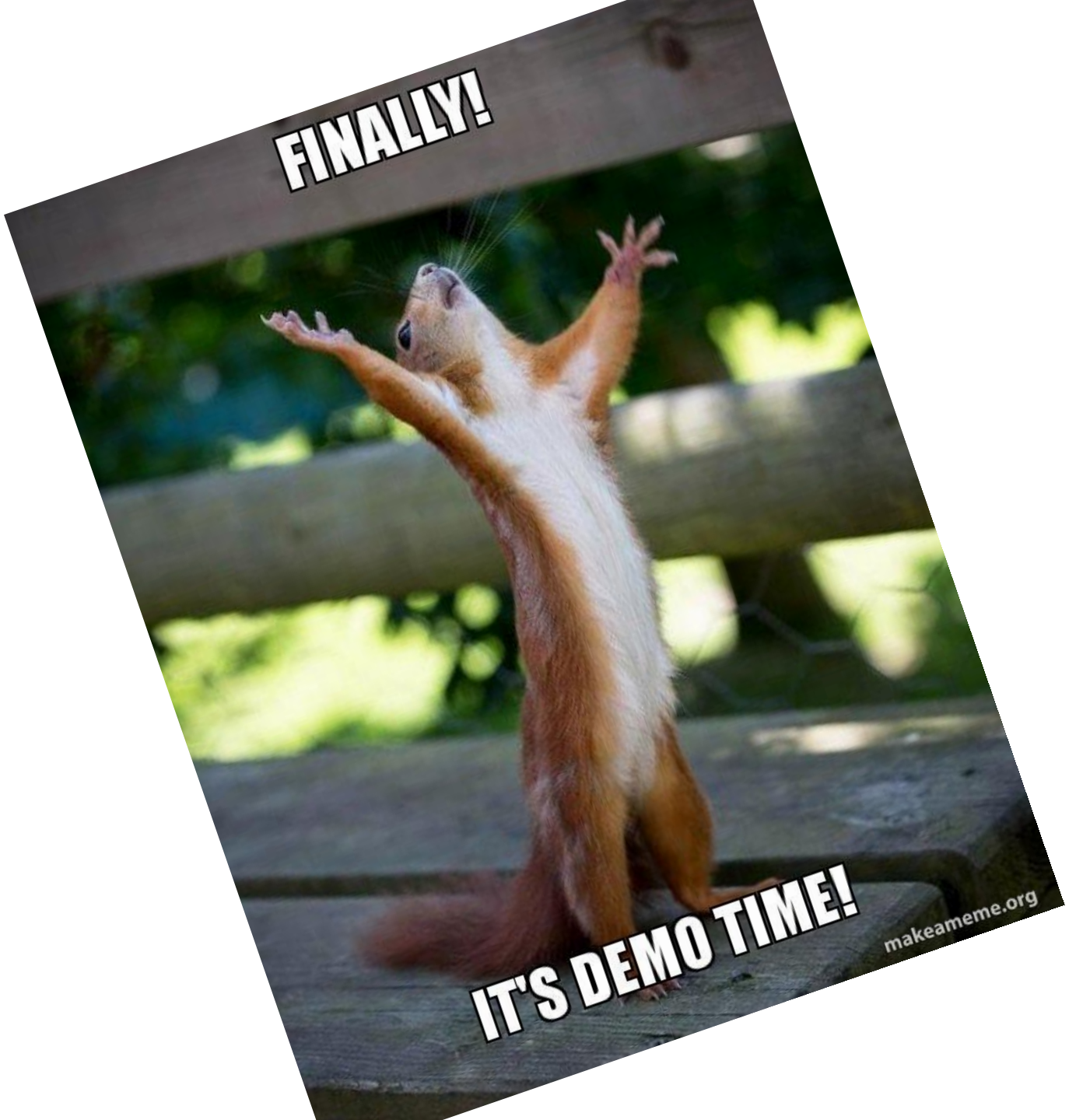
- "Walmart Teacher"
- Low-cost student activities to teach water and water treatment





- Coagulation jars
- Table top filters
- Portable microscopes
- Water test stripes
- Secchi disc creation
- Glucose meters for microbial activity (Beta Testing!)




FINALLY!
IT'S DEMO TIME!
makeameme.org

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